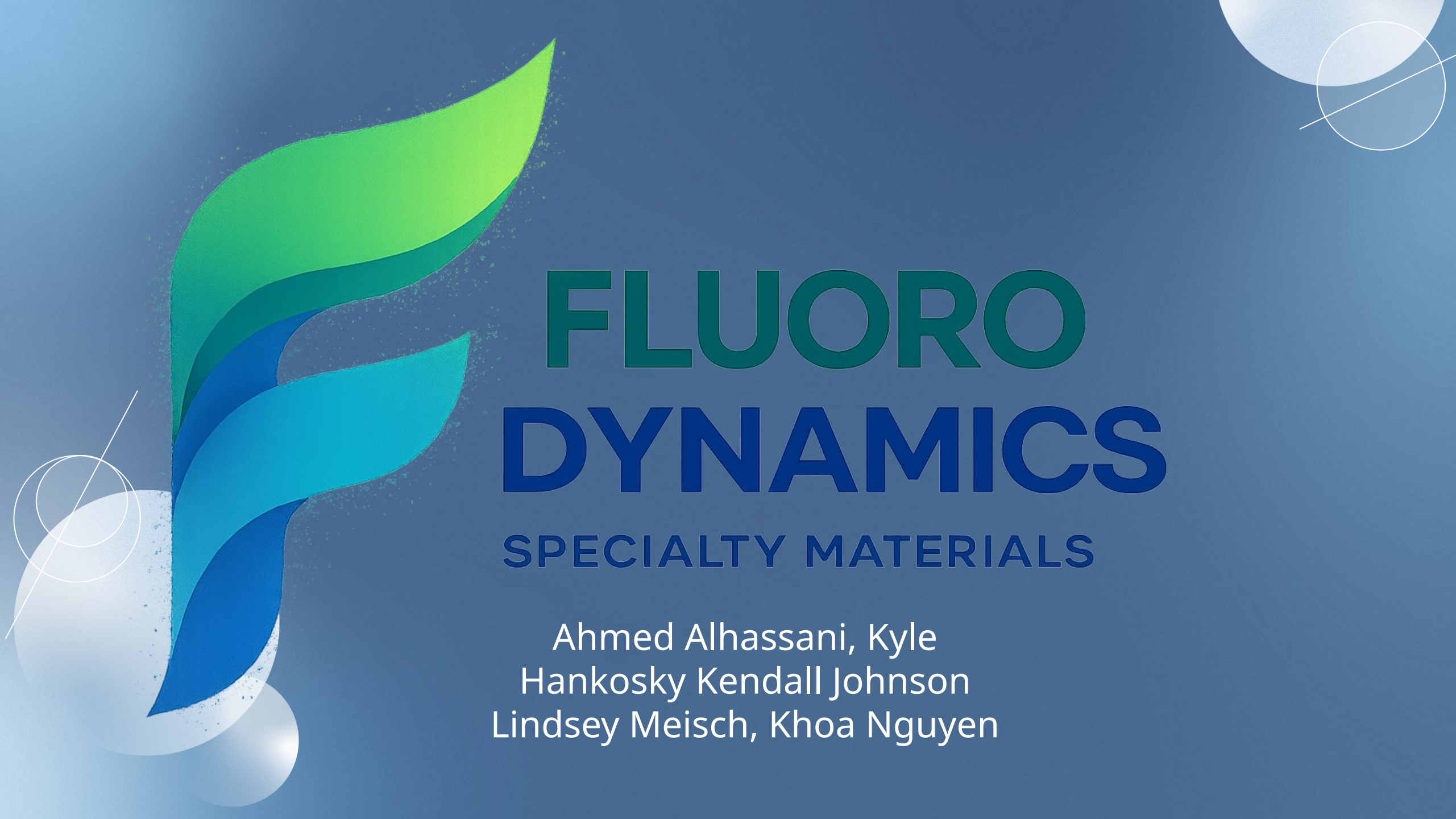




FLUORO DYNAMICS

SPECIALTY MATERIALS

Ahmed Alhassani, Kyle
Hankosky Kendall Johnson
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PFD and
Material Balance

03

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04

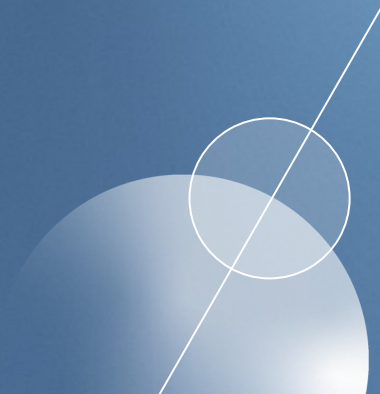
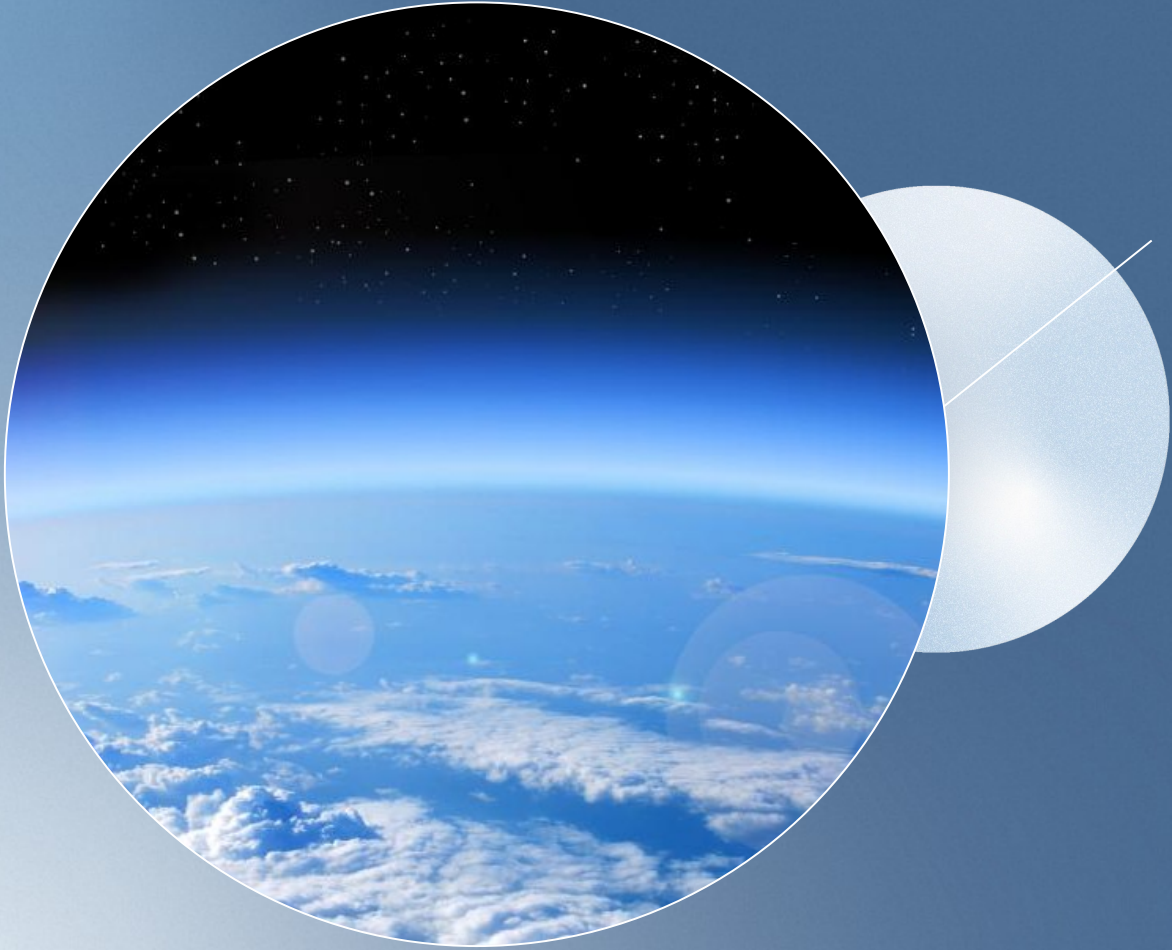
Equipment
Sizing and List

05

References

01

Introduction



About HFO-1234yf

4

What It Is

- A next generation hydrofluoroolefin (HFO) refrigerant
- Replacement for HFC-134a in automotive A/C systems
- Low GWP (<1 to 4)

The Problem

- HFC-134a has a very high GWP (~1,430)
- International policies are requiring the phasing out of high-GWP refrigerants
- Automotive manufacturers need a low-GWP refrigerant that has the same performance

Design Basis

Manufacturing location: Michigan, US

Expected cars sold US 2025: 16.1-16.2 MM

Approx. Charge: 1 lb 1234yf/car

Supply Approx: 3 MM cars/year

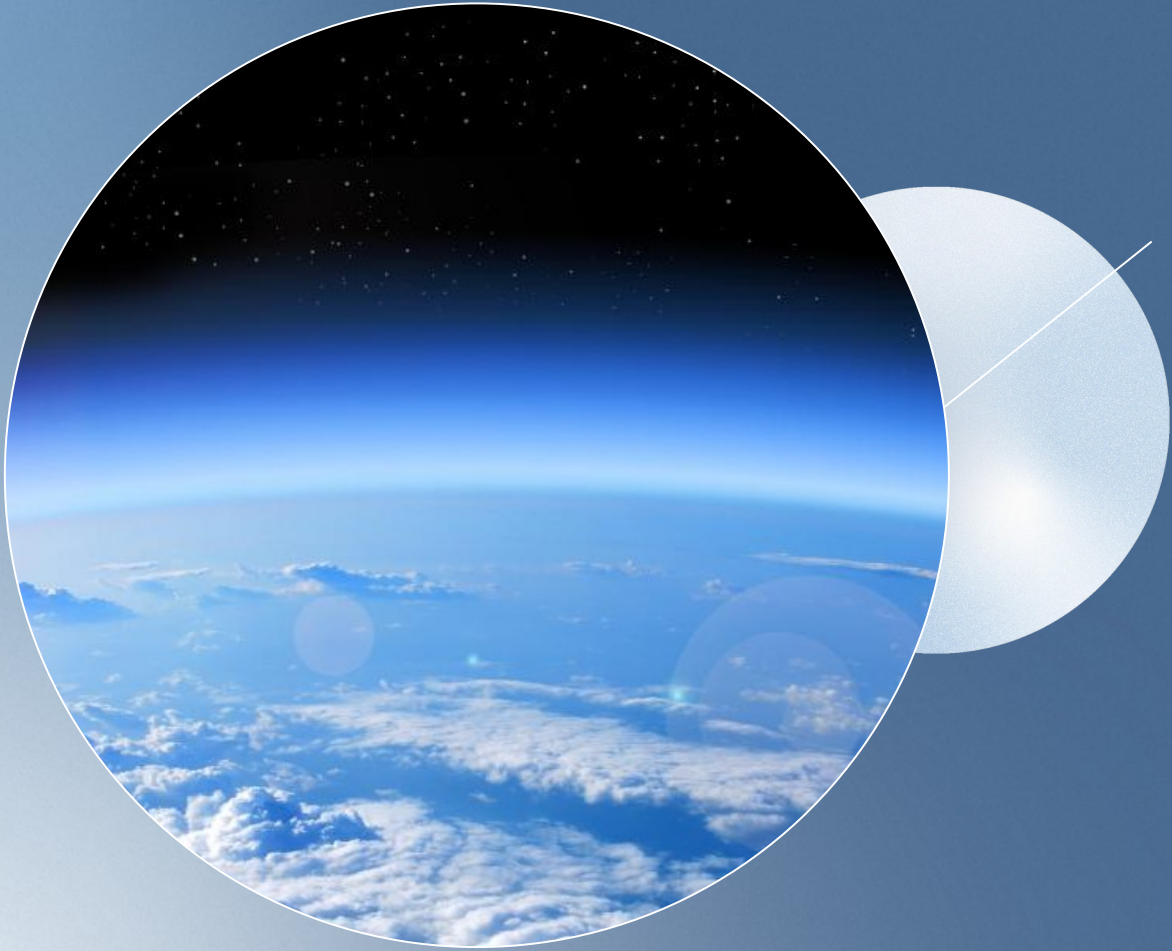
Need: 3 MM lb 1234yf/year



"Michigan County Map." **Wikimedia Commons**, Creative Commons Attribution–ShareAlike 4.0, https://commons.wikimedia.org/wiki/File:Michigan_County_Map.svg

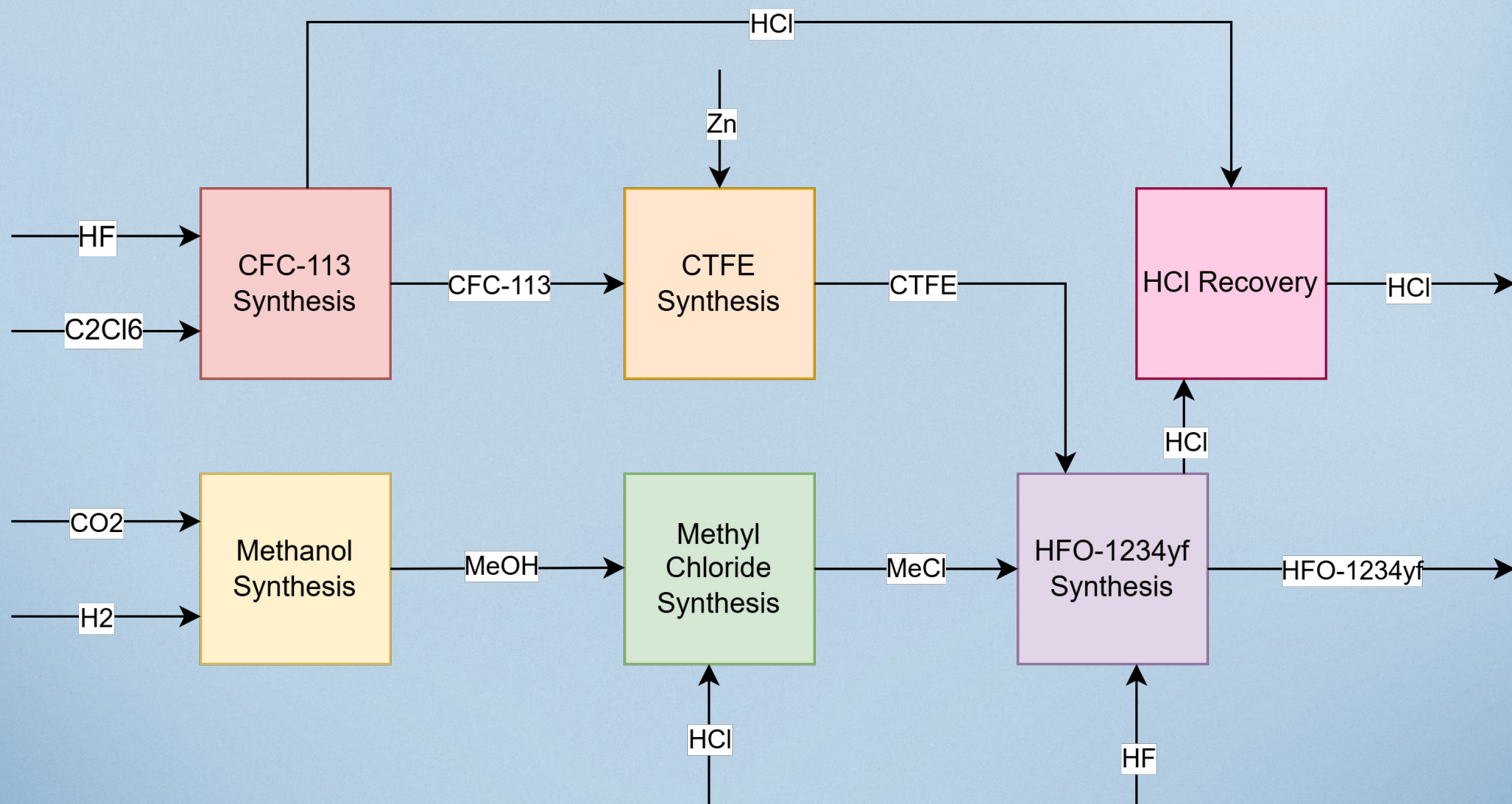
02

PFD and Material Balance



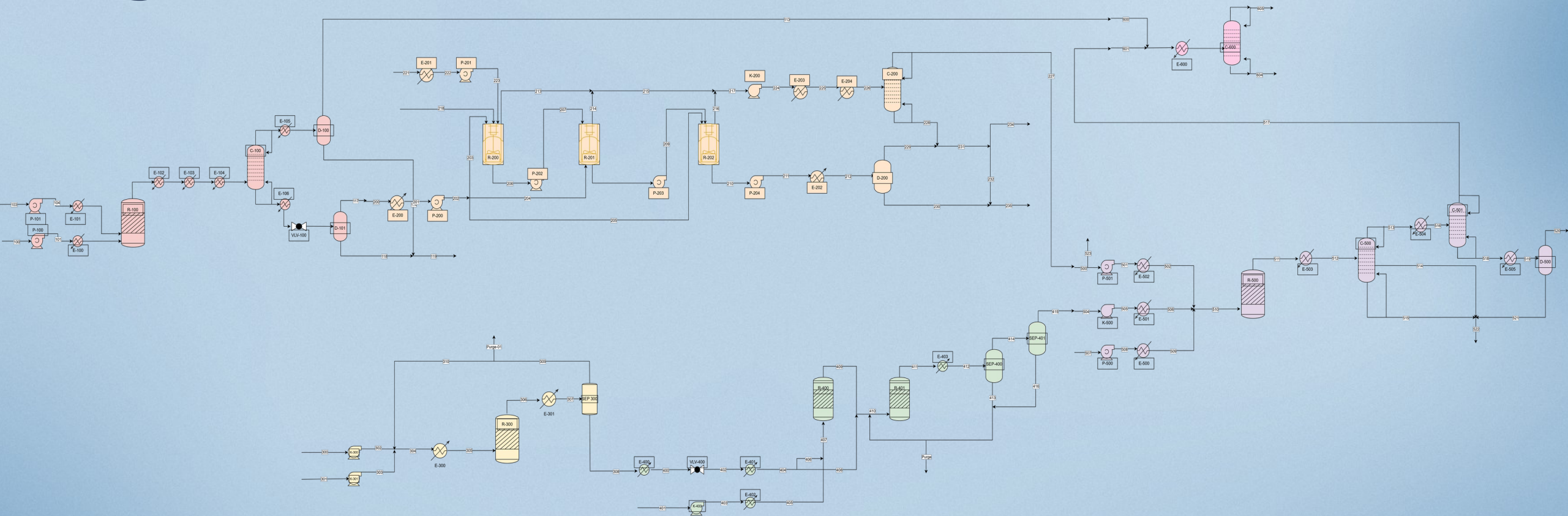
Block Flow Diagram

7



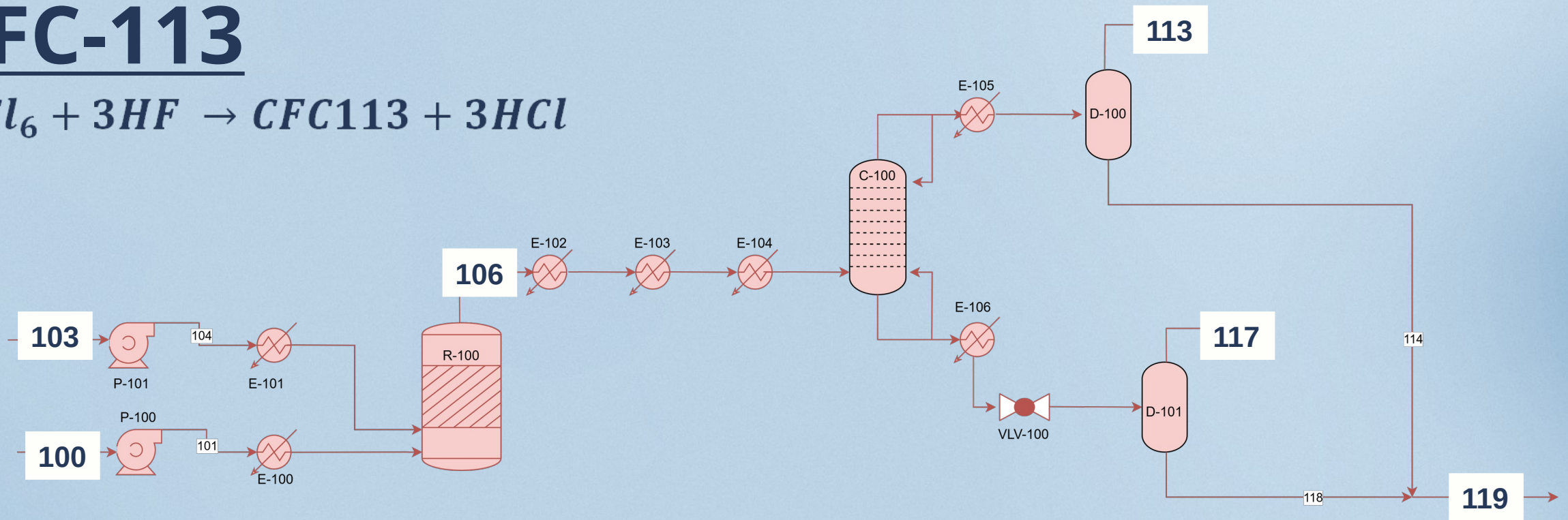
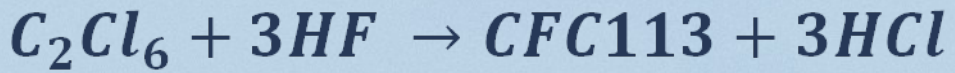
Process Flow Diagram

8

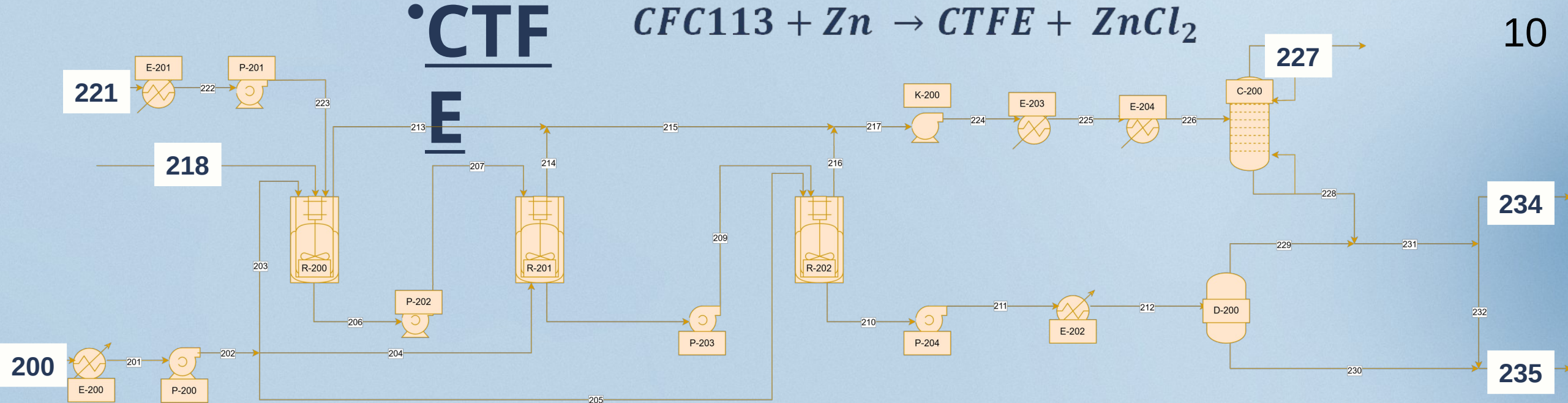


CFC-113

9

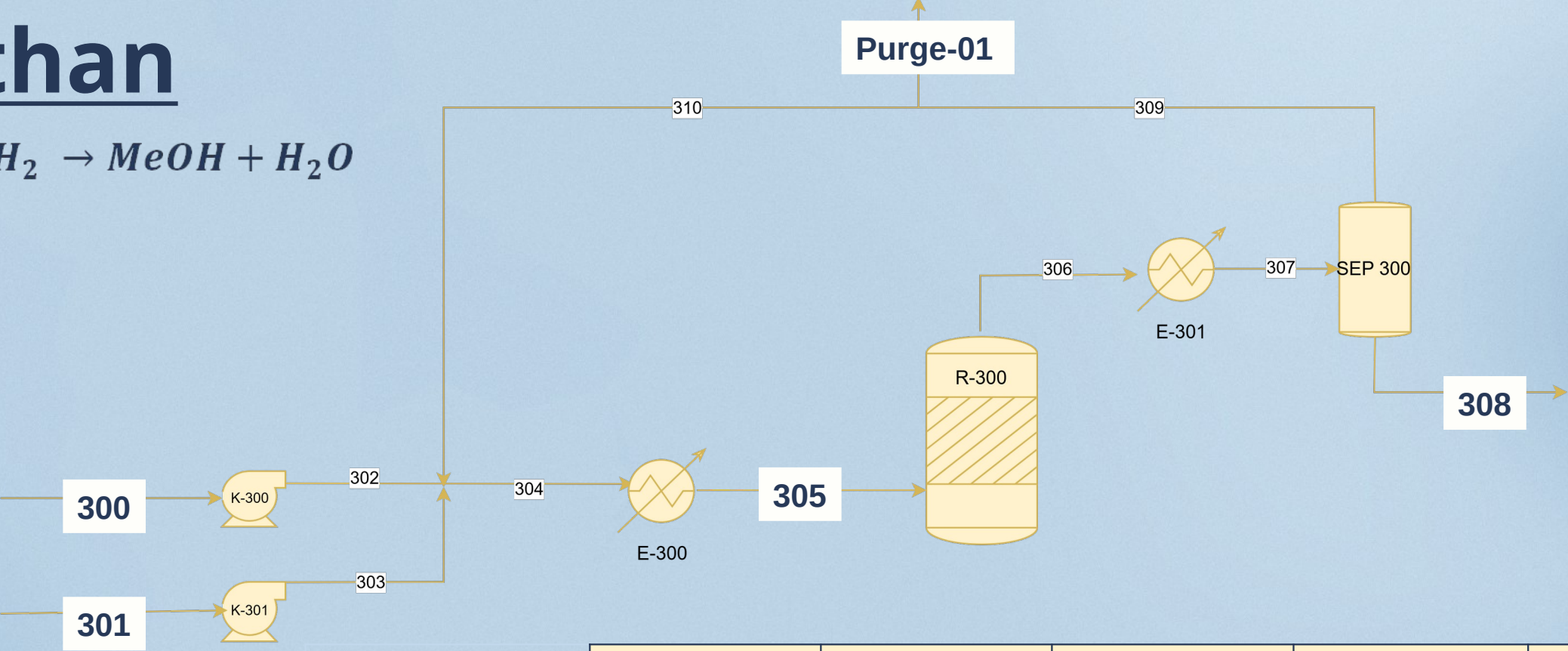
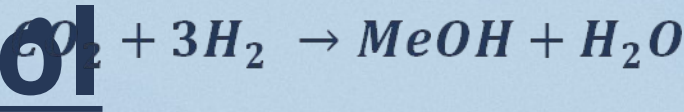


	100	103	106	113	117	119
Temperature (F)	77	50	910	-4	197	-4
Pressure (psia)	15	15	71	57	19	57
	Mass Flow (lb/hr)					
Mass Flow	1,113	282	1396	518	877	0.1
Hydrogen Fluoride	-	282	6	6	<<1	<1
Hydrogen Chloride	-	-	503	503	-	<1
Hexachloroethane	1,113	-	24	-	24	-
CFC-113	-	-	860	8	851	<1
Impurities	-	-	3	0.9	2.2	-



	200	218	221	227	234	235
Temperature (F)	197	77	95	32	146	148
Pressure (psia)	19	15	14	44	15	15
	Mass Flow (lb/hr)					
Mass Flows	877	331	940	514	940	767
Hydrogen Fluoride	<<1	-	<1	<<1	<1	<<1
Hexachloroethane	24	-	132	<<<1	132	24
CFC-113	851	-	<1	<<1	<1	<<1
Pentachlorofluoroethane	2	-	17	<<<1	17	2
Methanol	-	-	648	<1	648	73
Zinc	-	331	-	-	-	34
Zinc Chloride	-	-	-	-	-	619
CTFE	-	-	142	514	142	16
Impurities	2.3	-	1.5	<<1	1.5	0.2

Methan

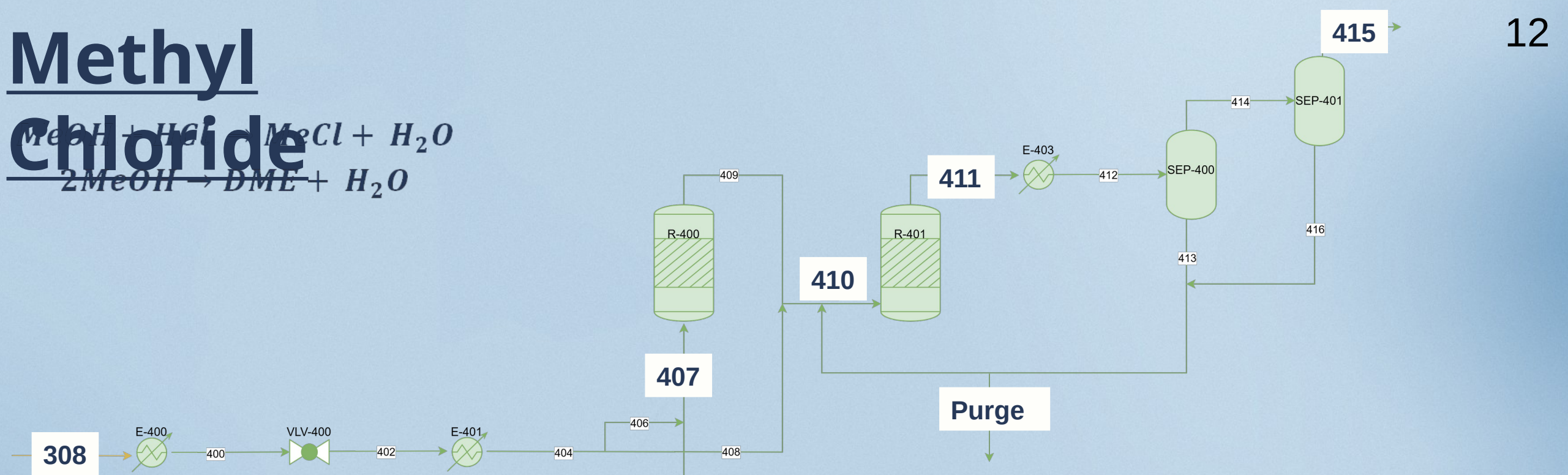


	300	301	305	308	Purge - 01
Temperature (F)	77	77	401	140	140
Pressure (psia)	15	15	725	725	725
	Mass Flow (lb/hr)				
Mass Flow	2,859	30	26,865	225	2,664
Methanol	-	-	308	124	34
Water	-	-	42	84	5
Carbon Dioxide	2,859	-	26,485	16	2,625
Hydrogen	-	30	30	-	-

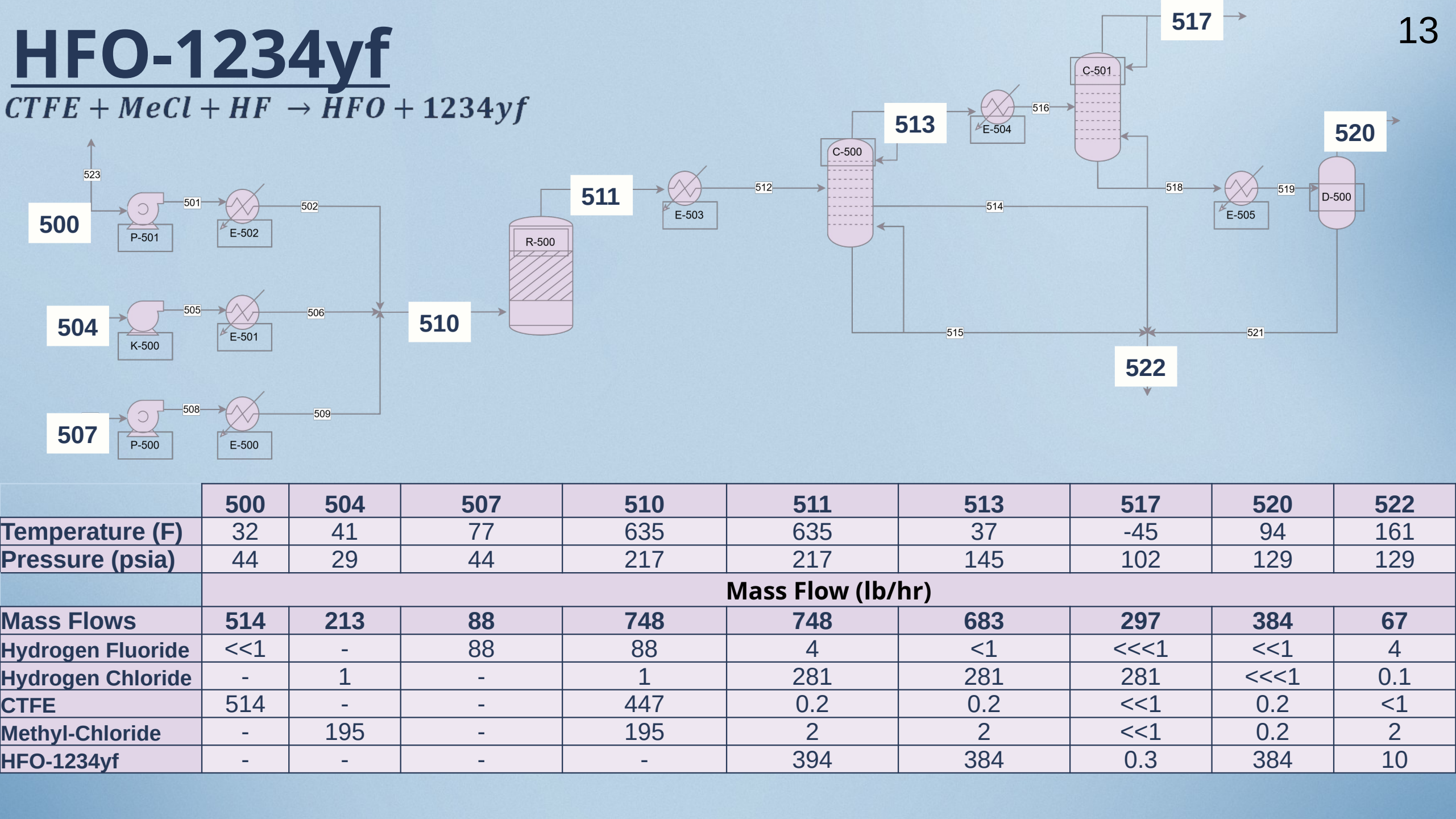
Methyl



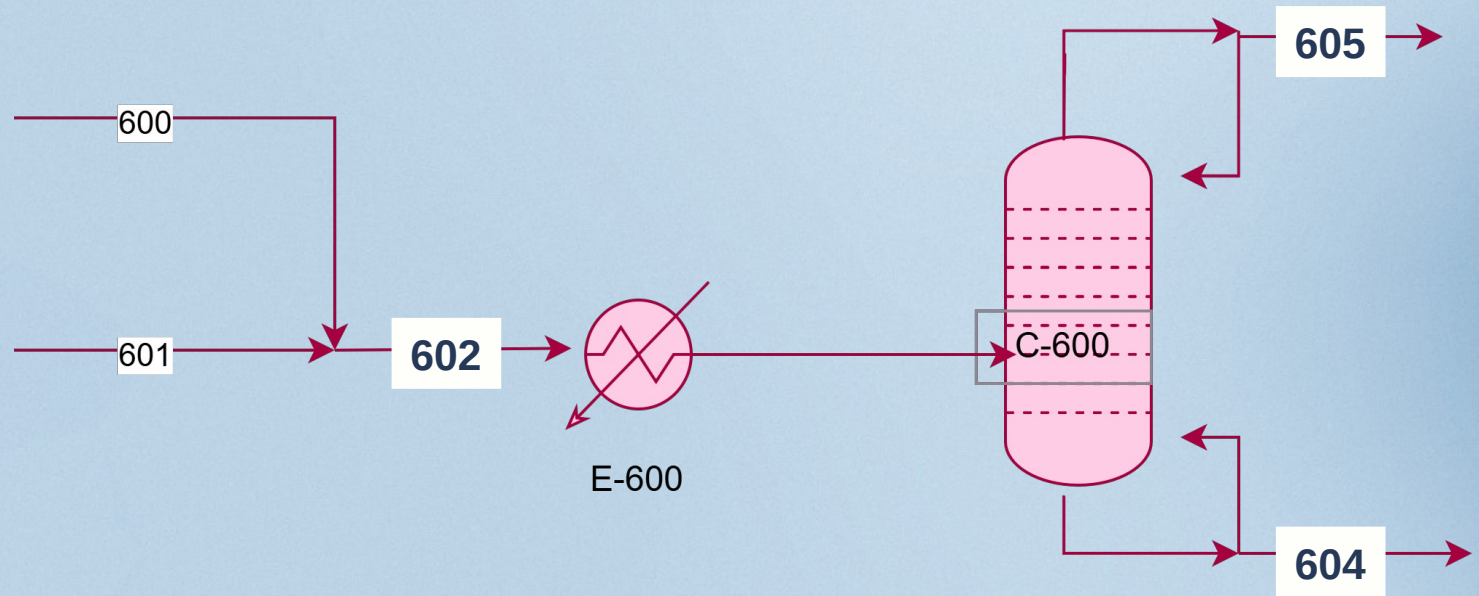
12



	308	401	407	410	411	415	Purge
Temperature (F)	140	77	244	188	302	41	104
Pressure (psia)	725	15	28	28	29	29	29
	Mass Flow (lb/hr)						
Mass Flows	225	142	300	1,734	1,734	213	377
Hydrogen Chloride	-	142	142	45	1	1	<1
Methanol	124	-	87	40	1	<1	124
Methyl-Chloride	-	-	-	135	197	195	<1
Water	84	-	59	1,500	1,500	<1	236
Carbon Dioxide	16	-	11	16	16	16	16
DME	-	-	-	<1	<1	<1	-



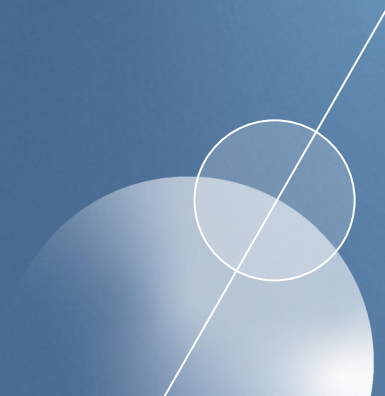
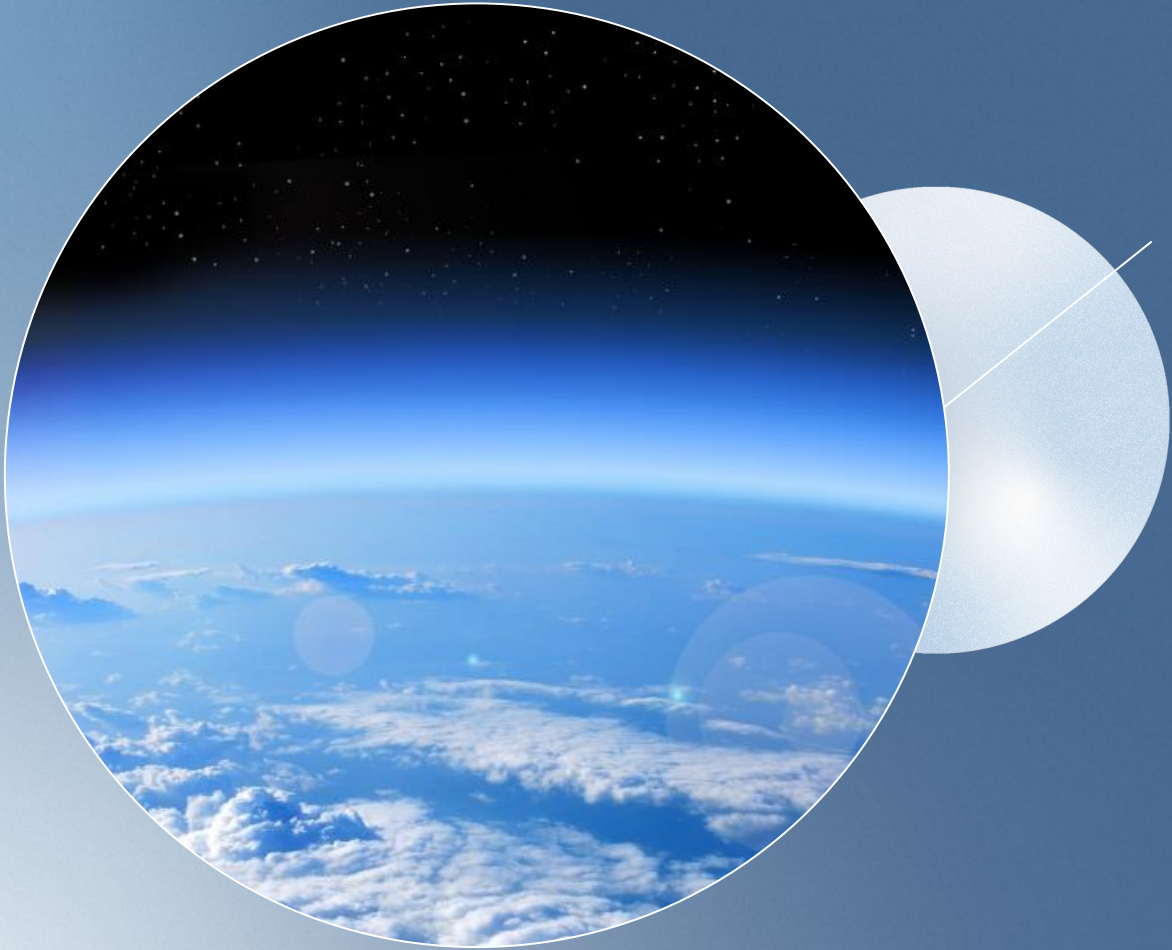
HCl Recovery



	602	604	605
Temperature (F)	-19	-53	-71
Pressure (psia)	57	57	57
	Mass Flow (lb/hr)		
Mass Flows	815	43	772
Hydrogen Fluoride	6	6	<<<1
Hydrogen Chloride	784	28	756
1,2,2-Trichloro-1,1,2-Trifluoroe	8	8	<<<1
Carbon Dioxide	16	<1	16

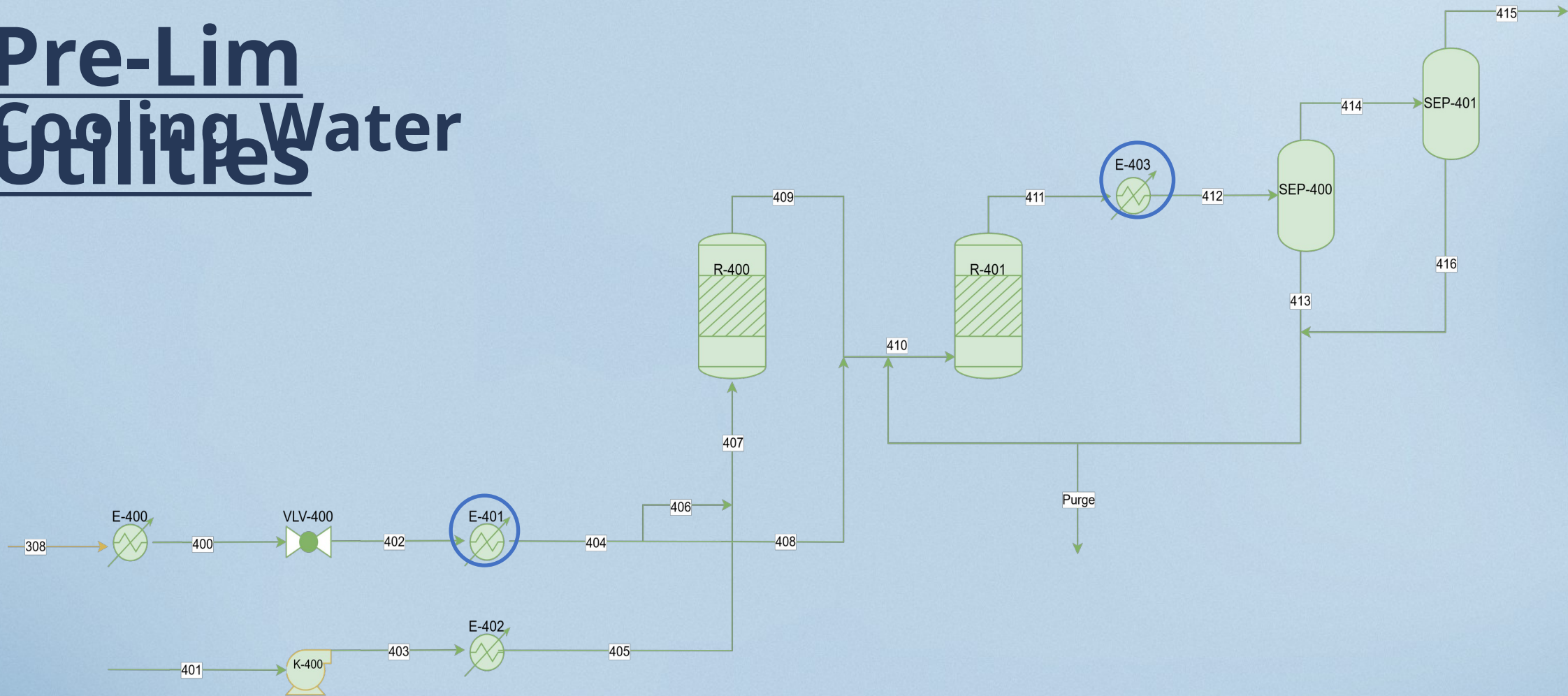
03

Preliminary Utilities



Pre-Lim Cooling Water Utilities

16

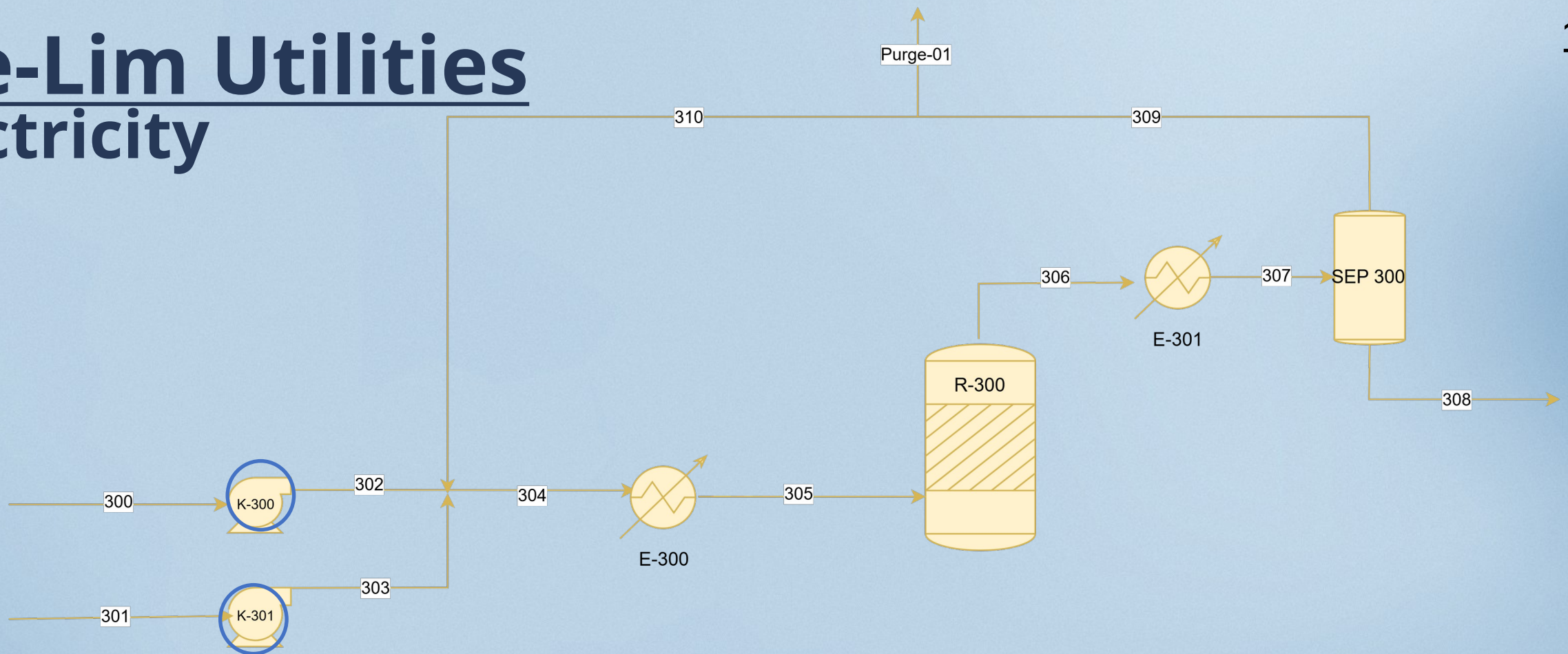


Equipment	Usage Value	Usage Units	Cost (\$/hr)
E-401	2.02	gpm	0.17
E-403	395.42	gpm	33.45
Total	397.44	gpm	33.62

Pre-Lim Utilities

Electricity

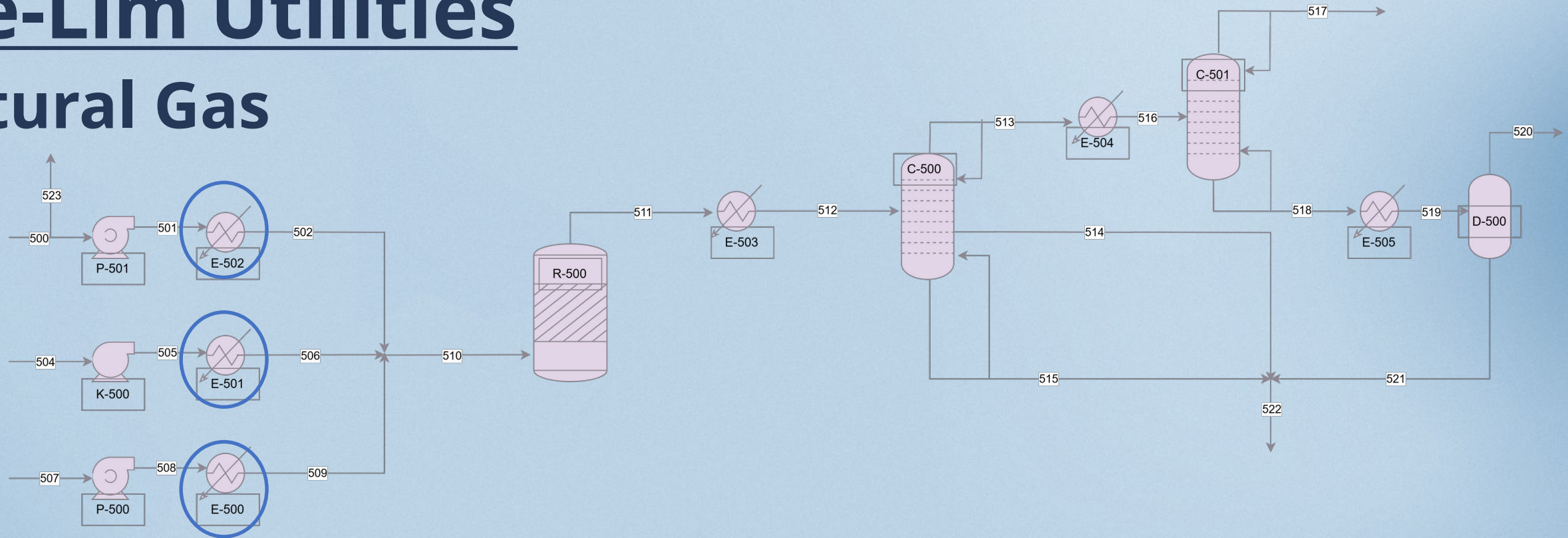
17



Equipment	Usage Value	Usage Units	Cost (\$/hr)
K-300	141.60	kW	19.50
K-301	39.31	kW	5.41
Total	180.91	kW	24.91

Pre-Lim Utilities

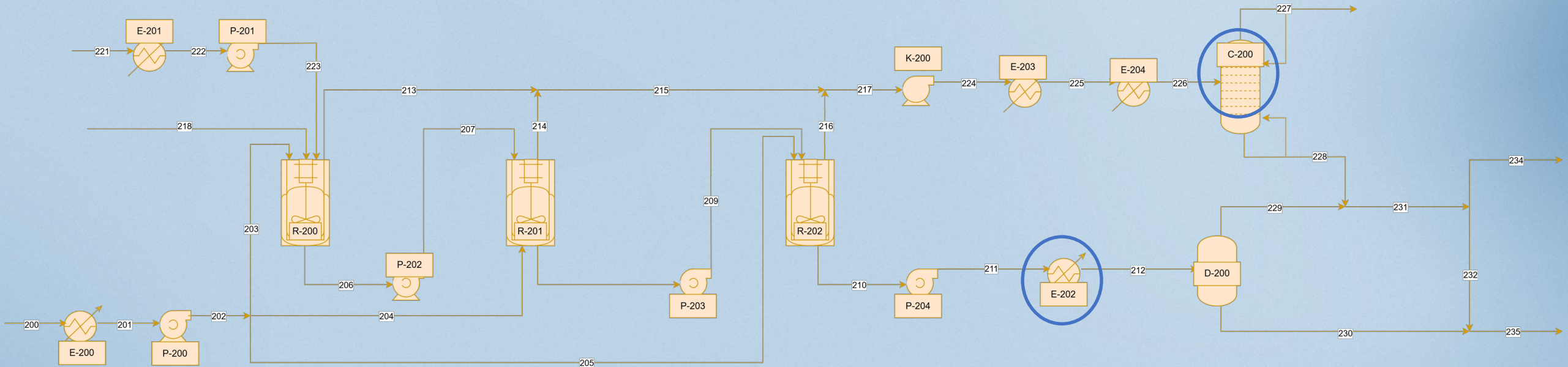
Natural Gas



Equipment	Usage Value	Usage Units	Cost (\$/hr)
E-500	85,309.12	Btu/hr	0.32
E-501	17,464.08	Btu/hr	0.06
E-502	71,605.73	Btu/hr	0.26
Total	174,378.93	Btu/hr	0.64

Pre-Lim Utilities

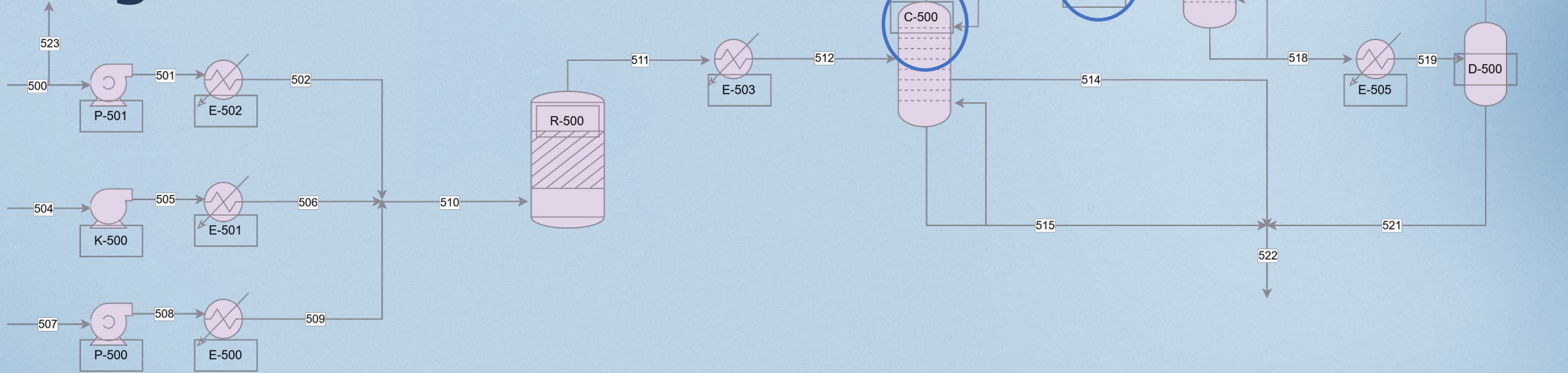
Steam



Equipment	Usage Value	Usage Units	Cost (\$/hr)
C-200	106.32	lb/hr	4.79
E-202	353.27	lb/hr	15.91
Total	459.59	lb/hr	20.7

Pre-Lim Utilities

Refrigeration



Equipment	Usage Value	Usage Units	Cost (\$/hr)
C-500	5.22	ton	0.25
C-501	1.02	ton	0.21
E-504	0.84	ton	0.13
Total	7.08	ton	0.59

Cooling
Water

Equipment	Usage Value	Usage Units	Cost (\$/hr)
E-104	9.75	gpm	0.82
E-200	14.16	gpm	1.20
E-201	63.78	gpm	5.39
E-203	22.10	gpm	1.87
E-301	489.55	gpm	41.41
E-401	2.02	gpm	0.17
E-403	395.42	gpm	33.45
E-503	24.07	gpm	2.04
Total	1,020.85	gpm	86.35

Natural
Gas

Equipment	Usage Value	Usage Units	Cost (\$/hr)
E-500	85,309.12	Btu/hr	0.32
E-501	17,464.08	Btu/hr	0.06
E-502	71,605.73	Btu/hr	0.26
E-100	88,262.62	Btu/hr	0.33
E-101	260,232.29	Btu/hr	0.96
Total	522,873.83	Btu/hr	1.93

Steam			
Equipment	Usage Value	Usage Units	Cost (\$/hr)
C-100 (Reboiler)	602.50	lb/hr	27.13
C-200 (Reboiler)	106.32	lb/hr	4.79
C-500 (Reboiler)	274.98	lb/hr	12.38
E-106	49.39	lb/hr	2.22
E-202	353.27	lb/hr	15.91
E-300	1,777.28	lb/hr	61.83
E-400	224.40	lb/hr	7.81
E-402	3.22	lb/hr	0.11
E-505	24.76	lb/hr	1.11
Total	3,416.12	lb/hr	133.29

Refrigeration			
Equipment	Usage Value	Usage Units	Cost (\$/hr)
C-100 (Condenser)	13.87	ton	1.57
C-200 (Condenser)	2.19	ton	0.16
C-500 (Condenser)	5.22	ton	0.25
C-501 (Condenser)	1.02	ton	0.21
C-600 (Condenser)	3.62	ton	0.56
E-204	0.89	ton	0.09
E-504	0.84	ton	0.13
E-600	0.78	ton	0.23
Total	28.43	ton	3.18

Electricit

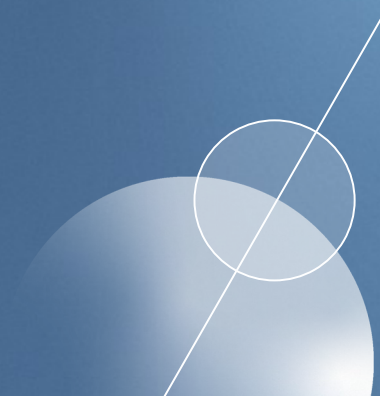
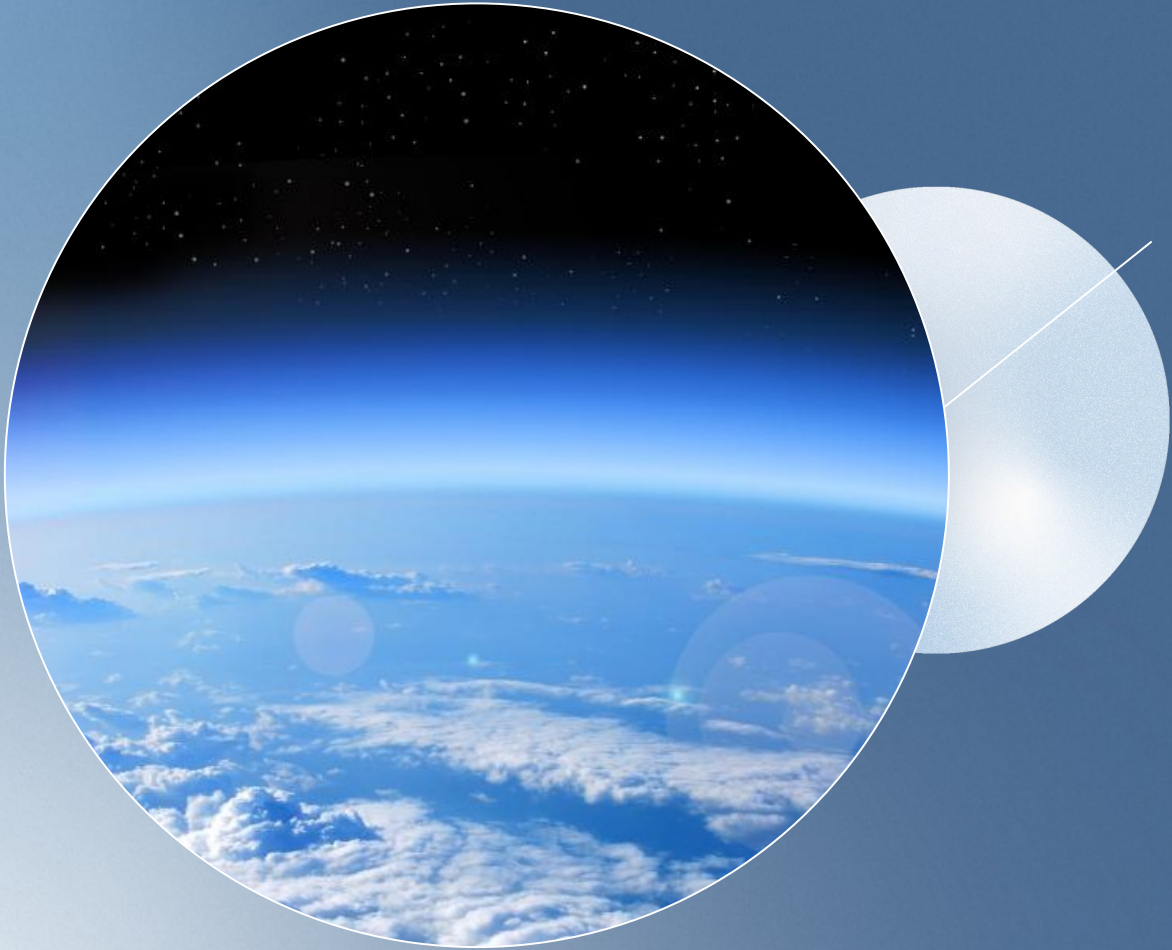
Equipment	Usage Value	Usage Units	Cost (\$/hr)
P-501	0.08	kW	0.01
C-501 (Reboiler)	20.53	kW	2.83
C-600 (Reboiler)	59.67	kW	8.22
E-105	30.17	kW	4.15
K-200	8.78	kW	1.21
K-300	141.60	kW	19.50
K-301	39.31	kW	5.41
K-400	1.09	kW	0.15
K-500	4.60	kW	0.63
P-100	0.10	kW	0.01
P-101	0.05	kW	0.01
P-200	0.07	kW	0.01
P-202	0.02	kW	0.00
P-203	0.02	kW	0.00
P-204	0.02	kW	0.00
P-500	0.002	kW	0.00
Total	306.13	kW	42.15

Total Pre-Lim Utilities

Utility	\$/hr
Electricity	42.15
Cooling Water	86.35
Natural Gas	1.93
Steam	133.29
Refrigeration	3.18
Total	266.9
Total Utility Cost Per Year	\$2MM

04

Equipment List and Sizing



Hand Calculation

	Hand Calc	Aspen
Column Trays	37 trays	39 trays
Column Diameter	4.60 ft	1.50 ft
Column Height	77.0 ft	90.0 ft

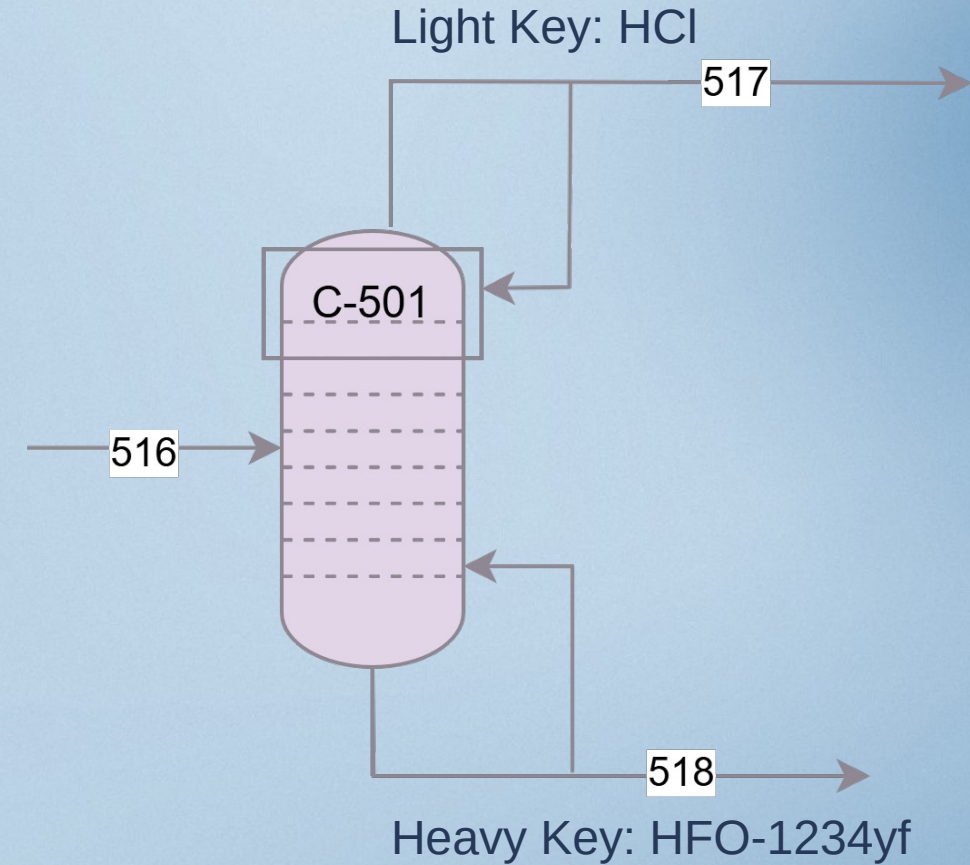
$$N_{min} = \frac{\log\left(\frac{x_{D,LK}}{x_{D,HK}} * \frac{x_{B,HK}}{x_{B,LK}}\right)}{\log(\alpha)}$$

$$N = \frac{N_{min}}{\eta}$$

$$u_{flood} = C \sqrt{\frac{\rho_L - \rho_v}{\rho_v}}$$

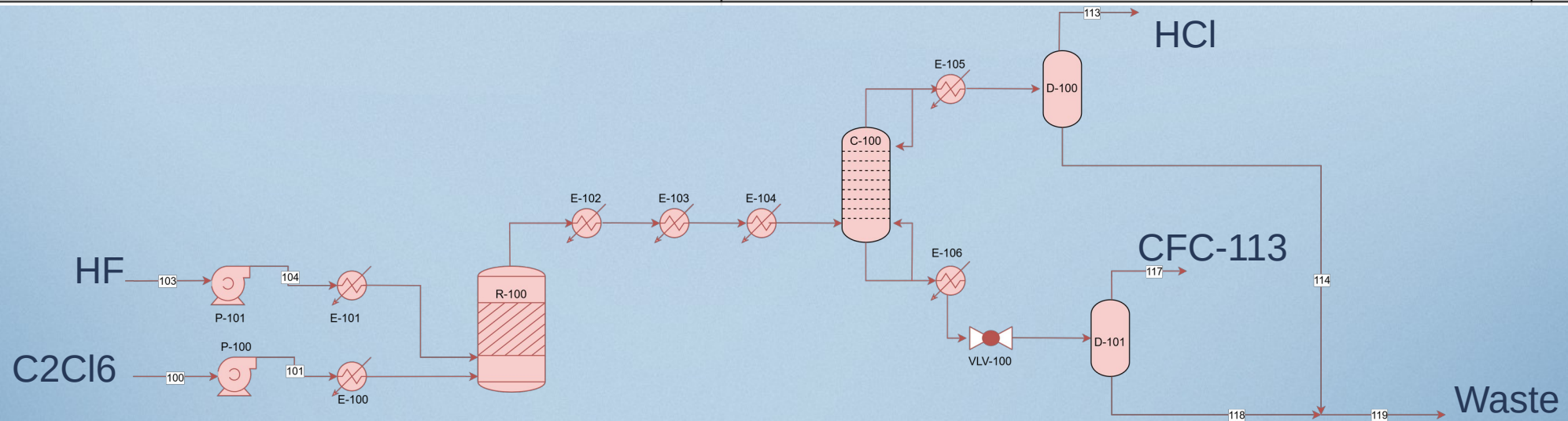
$$A = \frac{\dot{Q}}{0.7 * u_{flood}}$$

$$D = \sqrt{\frac{4A}{\pi}}$$



Equipment List/Sizing/MOC

Tag	Type	Size	Material
P-100	Centrifugal pump	Flow Rate = 1.25 gpm	Carbon Steel
E-100	Shell-and-tube heat exchanger	A = 5 sqft	Carbon Steel
P-101	Centrifugal pump	Flow Rate = 0.62 gpm	Monel 400
E-101	Shell-and-tube heat exchanger	A = 30 sqft	Monel 400
R-100	Fixed-bed reactor	Volume = 38 gal; Diameter = 1 ft; Height = 6.5 ft	Monel 400
E-102	Shell-and-tube heat exchanger	A = 3 sqft	Monel 400
C-100	Tray distillation column (sieve trays)	33 trays; Diameter = 1.50 ft; Height = 78 ft; Tray Spacing = 2 ft	Monel 400
E-103	Shell-and-tube heat exchanger	A = 5 sqft	Monel 400
D-100	Vertical vessel	Volume = 620 gal; Diameter = 3 ft; Height = 12 ft	Hastealloy
E-104	Shell-and-tube heat exchanger	A = 5.22 sqft	Monel 400
D-101	Verticle vessel	Volume = 620 gal; Diameter = 3 ft; Height = 12 ft	Carbon Steel
E-105	Shell-and-tube condenser	A = 2.03 sqft	Hastealloy
E-106	Shell-and-tube heat exchanger	A = 2.88 sqft	Carbon Steel
VLV-100	Globe control valve	-	Carbon Steel

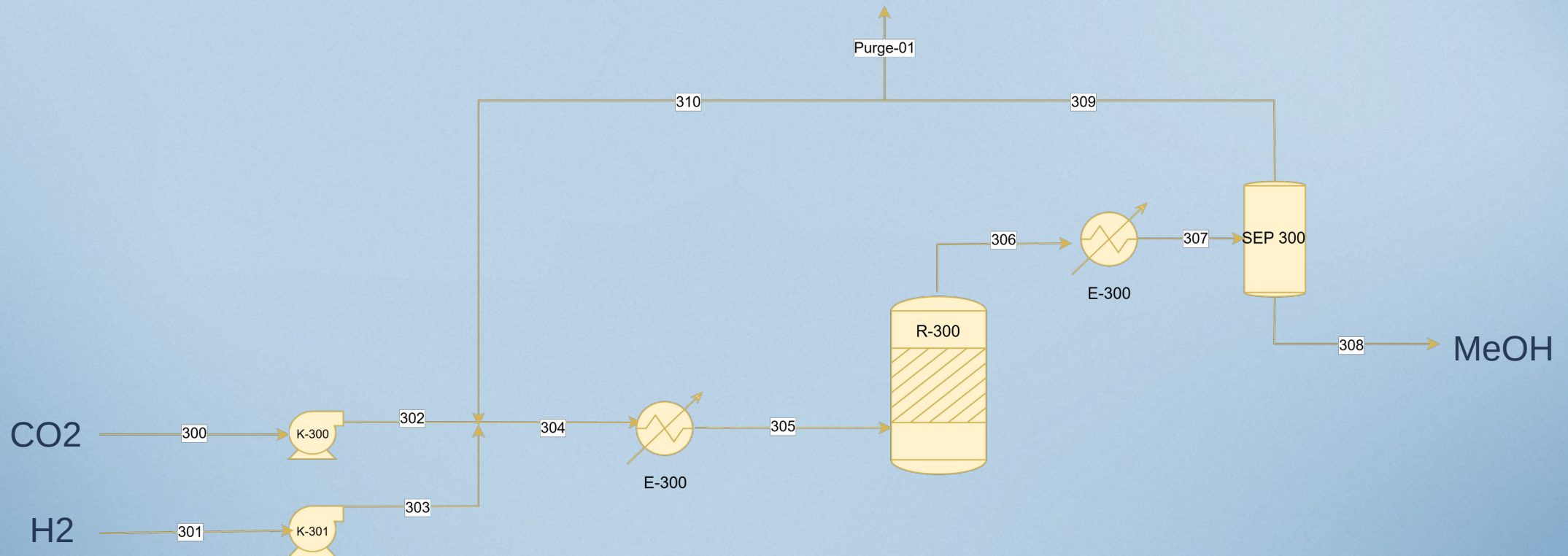


The diagram illustrates the chemical synthesis process for CTFE. It begins with the input of MeOH (stream 221) and Zn (stream 218). MeOH is heated in exchanger E-201 and pumped by P-201 to reactor R-200. Zn is added to R-200. The reaction mixture is pumped by P-202 to reactor R-201, where it is heated by exchanger E-203. The mixture then moves to reactor R-202, where it is heated by exchanger E-204. The final product, CTFE, is collected in stream 227. The waste stream is labeled 'Waste' and includes streams 234, 232, and 235.

Equipment

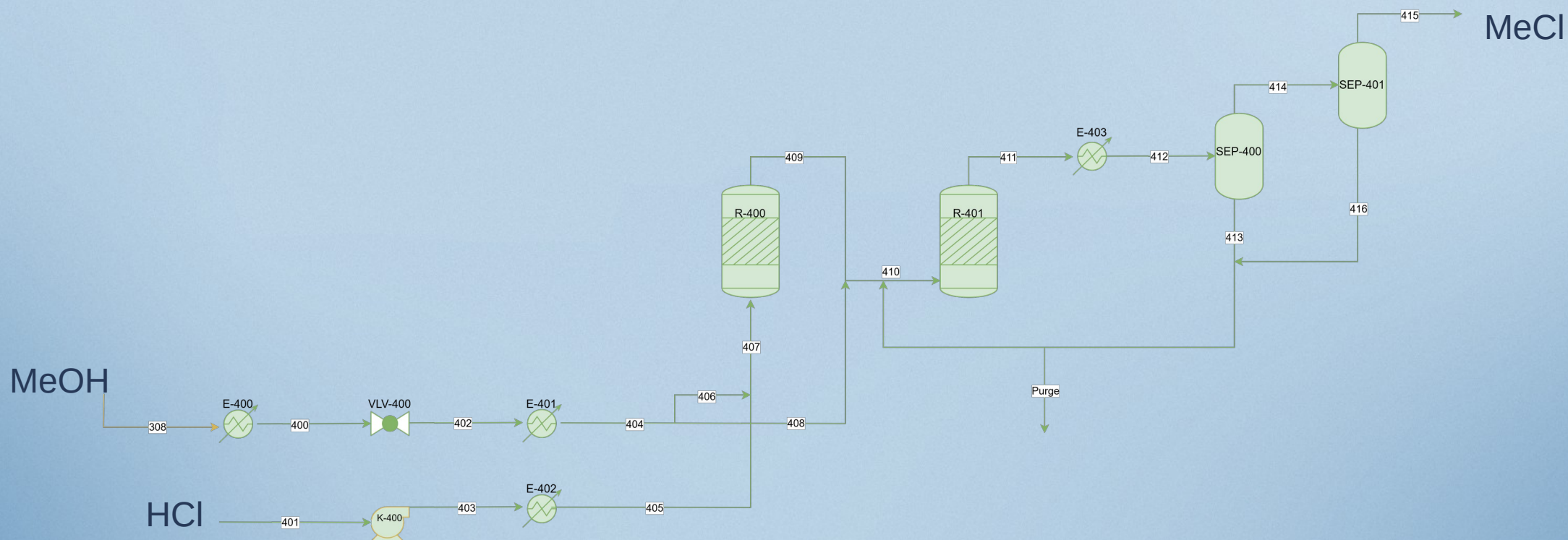
Process Flow Diagram (PFD)

Tag	Type	Size	Material
K-300	Centrifugal gas compressor	Flow = 2,150 gpm; Power = 93 kW	Carbon Steel
K-301	Centrifugal gas compressor	Flow = 71 gpm; Power = 4 kW	Carbon Steel
E-300	Shell-and-tube heat exchanger	A = 169.64 sqft	Carbon Steel
R-300	Plug flow reactor	Diameter = 0.05 ft; Height = 5.5 ft	Carbon Steel
E-301	Shell-and-tube heat exchanger	A = 64 sqft	Carbon Steel
SEP-300	Vertical vessel	Volume = 620 gal; Diameter = 3 ft; Height = 12 ft	Carbon Steel



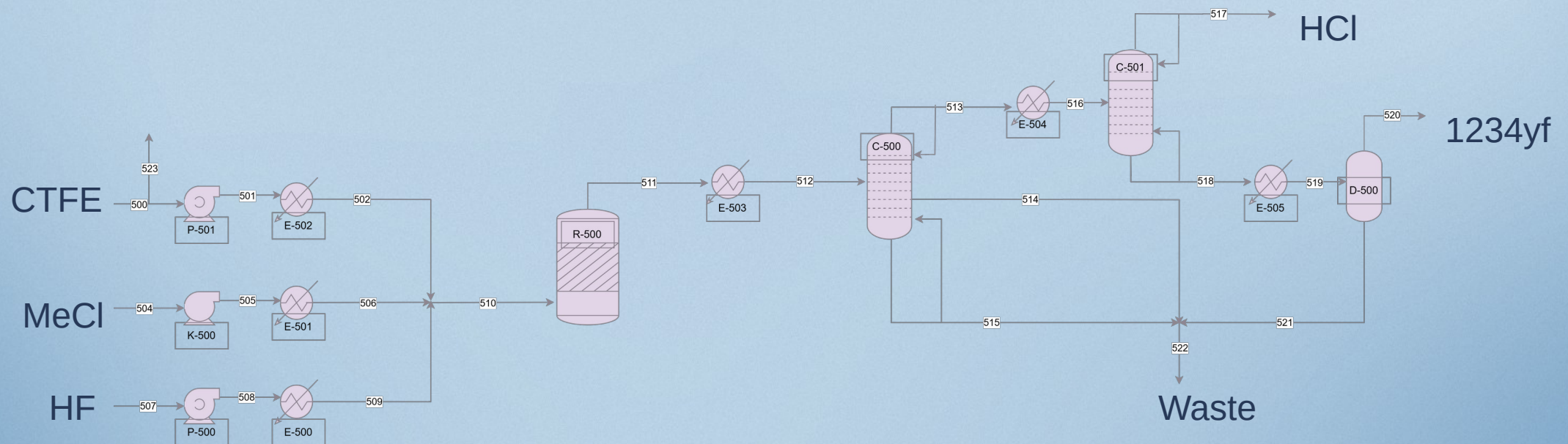
Equipment List/Sizing/MOC

Tag	Type	Size	Material
E-400	Shell-and-tube heat exchanger	A = 9.92 sqft	Carbon Steel
VLV-400	Globe control valve	-	Carbon Steel
K-400	Centrifugal gas compressor	Flow = 209 gpm; Power = 1.09 kW	Hastealloy
E-401	Shell-and-tube heat exchanger	A = 0.49 sqft	Carbon Steel
R-400	Fixed-bed reactor	Volume = 41 gal; Diameter = 1 ft; Height = 7 ft	Hastealloy
R-401	Fixed-bed reactor	Volume = 41 gal; Diameter = 1 ft; Height = 7 ft	Hastealloy
E-402	Shell-and-tube heat exchanger	A = 1.66 sqft	Hastealloy
E-403	Shell-and-tube heat exchanger	A = 178 sqft	Hastealloy
SEP-400	Vertical vessel	Volume = 620 gal; Diameter = 3 ft; Height = 12 ft	Hastealloy



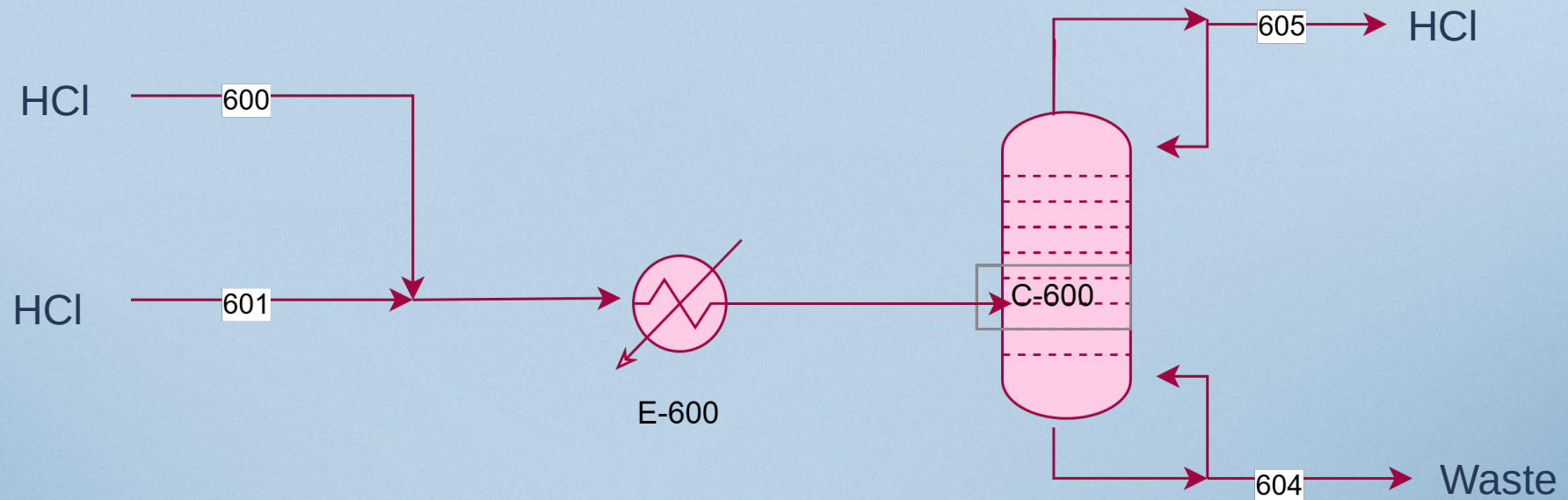
Equipment List/Sizing/MOC

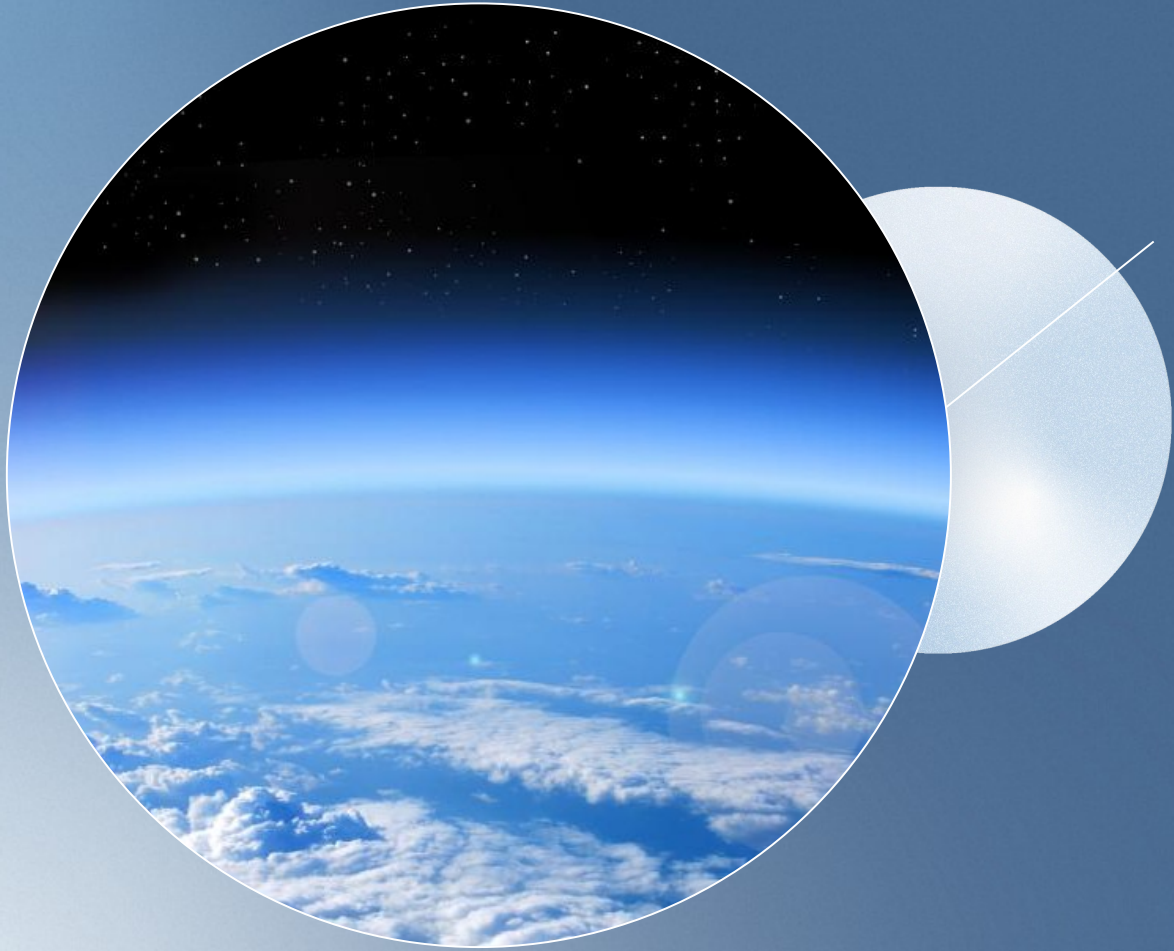
Tag	Type	Size	Material
P-500	Centrifugal pump	Flow Rate = 0.21 gpm	Monel 400
P-501	Centrifugal pump	Flow = 1.22 gpm	Carbon Steel
K-500	Centrifugal gas compressor	Flow = 105 gpm; Power = 4.60 kW	Hastealloy
E-500	Shell-and-tube heat exchanger	A = 5 sqft	Monel 400
E-501	Shell-and-tube heat exchanger	A = 2 sqft	Hastealloy
E-502	Shell-and-tube heat exchanger	A = 5 sqft	Carbon Steel
E-503	Shell-and-tube heat exchanger	A = 12 sqft	Monel 400
E-504	Shell-and-tube heat exchanger	A = 13 sqft	Hastealloy
R-500	Fixed-bed reactor (vertical packed bed)	A = 520 sqft	Monel 400
C-500	Tray distillation column (sieve trays)	33 Trays; Diameter = 1.5 ft; Height = 78 ft; Tray Spacing = 2 ft	Monel 400
C-501	Tray distillation column (sieve trays)	39 Trays; Diameter = 1.5 ft; Height = 90 ft; Tray Spacing = 2 ft	Hastealloy
D-500	Vertical vessel	Volume = 620 gal; Diameter = 3 ft; Height = 12 ft	Hastealloy



Equipment List/Sizing/MOC

Tag	Type	Size	Material
E-600	Shell-and-tube heat exchanger	0.39 sqft	Hastealloy
C-600	Tray distillation column	31 Trays; Diameter = 1.5 ft; Height = 74 ft; Tray Spacing = 2 ft	Hastealloy





05

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Thank
you!



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